

Raj Tharani

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EDUCATION

Madhav Institute of Technology & Science

Bachelor of Technology in Computer Science Engineering; GPA: 7.0

Gwalior, India

Sept 2022 – Jun 2026

SKILLS SUMMARY

- Languages:** Java, JavaScript, SQL, Python
- Full-Stack & Backend:** HTML, CSS, React.js, Node.js, Express.js, API Development, JWT, CI-CD
- Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Client-Server Architecture
- Databases & Tools:** MongoDB, PostgreSQL, Git, GitHub, Postman, Wireshark,
- Machine Learning / NLP:** Natural Language Processing (NLP), Transformers, Time-Series Modeling, Random Forest, Reinforcement Learning

WORK EXPERIENCE

Backend Developer Intern (AI/ML Systems) –Code4Bots Pvt Ltd | [🔗](#)

Jun 2025 – Sept 2025

- Designed a scalable text preprocessing pipeline for RAG-based coding queries, improving retrieval relevance by **~15%**.
- Standardized user inputs using query normalization and intent classification, reducing prompt variability and improving prompt consistency.
- Developed and fine-tuned a RoBERTa-based intent and category classifier** on large-scale coding queries as part of the ML backend pipeline.

Ethical Hacking & Cybersecurity Intern – CDAC Noida | [🔗](#)

Sept 2024 – Oct 2024

- Evaluated **10+** digital verification tools to detect phishing, spoofed emails, and manipulated media content.
- Identified and documented **12+** security vulnerabilities, proposing mitigation strategies to strengthen verification workflows.
- Analyzed threat patterns and verification outcomes to support internal security assessment and research improvements.

PROJECTS

MERN Food Delivery App (MERN Stack, Full-Stack) | [🔗](#)

Aug 2025 – Sept 2025

- Built and deployed a full-stack MERN food delivery application with **JWT-based authentication**, role-based access, and RESTful API architecture.
- Designed backend services using **Node.js and Express**, implemented **10+ RESTful API endpoints** for user, order, and authentication workflows.
- Implemented global state management using React Context API to manage **5+ application states**, including cart, authentication, and user sessions..
- Developed a **real-time analytics dashboard** to visualize orders, spending patterns, and user activity for improved user insights.

RoBERTa-Based Intent & Category Classifier (NLP, Transformers, PyTorch) | [🔗](#)

Jun 2025 – Sept 2025

- Trained a RoBERTa classifier on a **70k+** coding dataset, improving intent classification accuracy by **~15%**.
- Applied learning rate scheduling, warmup, and weight decay to reduce overfitting by **~20%** across **5 epochs**.
- Achieved stable, high-confidence inference using optimized hyperparameters and checkpoint selection.

Network Packet Traffic Analysis & Prediction (GRU + PPO) | [🔗](#)

Nov 2024 – Feb 2025

- Proposed an **AutoQoS framework** to address limitations of static QoS using ML-driven traffic prioritization across **20+ protocols**.
- Modeled protocol-level traffic behavior using **150MB Wireshark flow data** spanning **thousands of packets**.
- Implemented a PPO-based RL agent with **3 priority classes** (high/medium/low) for dynamic traffic scheduling.

CERTIFICATES

Full-Stack Web Development Bootcamp (Udemy) | [🔗](#)

Jul 2025

Completed a **62-hour** hands-on program building full-stack applications using React, Node.js, Express, and MongoDB.

Big Data Computing (NPTEL – IIT Kanpur) | [🔗](#)

Nov 2025

Completed an **8-week** IIT-led course on Hadoop and Spark pipelines; earned **Elite + Gold** certification with **91%**.

Data Structures & Algorithms in Java + 150 LeetCode Problems | [🔗](#)

Sept 2025

Solved **150+** algorithmic problems, strengthening core data structures and Java-based problem-solving skills

ACHIEVEMENT

First Position – Startup Ideathon (Dristikon Event) | [🔗](#)

Mar 2023

Demonstrated innovation and **entrepreneurial thinking** in idea presentation at Startup Cell MITS.

Third Position – Round Table Conference on Innovative Space Ideas | [🔗](#)

Nov 2023

Secured **3rd place** for proposing and discussing innovative space-related ideas in a **competitive academic forum**.

98.6 Percentile – JEE Main

Feb 2021

Achieved 98.6 percentile in the national-level engineering entrance examination among lakhs of candidates.